

IOT BASED PEST ALERTING SYSTEM

(Using Raspberry Pi 4B+)

A report submitted in partial fulfillment of the requirements for the award of

Bachelor of Technology

In

Electronics and Communication Engineering

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IoT Tools And Applications
(Skill Advanced Course)



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CERTIFICATE

This is to certify that the Mini Project titled "**IoT BASED PEST ALERTING SYSTEM USING RASPBERRY PI 4B+**" is a bonafide record of work done **G.S. ARAVIND REDDY (23BQ5A0427), V.THANUSH (23BQ5A0445), P.GOPINADH (23BQ5A0444), V.RAGHUVeer (22BQ1A04F3), R.NAVEEN (22BQ1A04C6)** as part of the Skill Advanced Course **IoT Tools And Applications** in partial fulfillment of the requirement of the degree for Bachelor of Technology in Electronics and Communication Engineering during the academic year 2025--2026.

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ABSTRACT

This project presents an innovative IoT-based pest alerting system designed to modernize agricultural pest management. The system utilizes a Raspberry Pi 4B+ as the central processing unit, integrated with a suite of sensors including a DHT11 for temperature and humidity monitoring, a YL-69 for soil moisture sensing, and a high-resolution web camera for image capture. The core of the system lies in its ability to perform real-time pest detection using advanced deep learning models like YOLOv5, which are deployed directly on the Raspberry Pi for on-edge analysis. This eliminates the dependency on constant manual inspection.

Environmental data from the sensors and image data from the camera are continuously collected and processed. When a pest is identified by the AI model, the system automatically triggers instant alerts to the farmer's smartphone and email. This allows for prompt and targeted intervention, significantly reducing the risk of widespread crop damage. All sensor readings and pest detection events are transmitted via the Pi's on-board Wi-Fi to a cloud platform, such as ThingSpeak or Firebase, where they are visualized on a user-friendly dashboard for remote monitoring.

By providing a cost-effective, automated, and scalable solution, this system helps in minimizing crop loss, reducing unnecessary pesticide use, and promoting sustainable farming practices. It empowers farmers with real-time intelligence, enabling precision agriculture and ensuring better crop health and yield.

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION OF PEST ALERTING SYSTEM:

- Pest infestation represents one of the most significant threats to global agriculture, leading to substantial crop damage, yield reduction, and economic losses for farmers. Traditional methods of pest control primarily involve manual scouting and scheduled pesticide application, which are not only labor-intensive but also often inefficient and reactive.
- These conventional approaches can result in delayed detection, allowing pests to proliferate and cause irreversible damage before action is taken. Furthermore, the over-reliance on chemical pesticides harms the environment, affects soil health, and poses risks to human health.
- The integration of Internet of Things (IoT) technology and Artificial Intelligence (AI) in agriculture offers a transformative solution. An IoT-based Pest Alerting System automates the entire process of pest monitoring and identification. By deploying a network of sensors and cameras in the field, the system continuously gathers crucial data on environmental conditions and visual cues of pest presence.
- This data is processed in real-time on edge devices like the Raspberry Pi 4B+, which runs sophisticated machine learning models for accurate pest detection. Upon identifying a threat, the system instantly notifies the farmer through mobile alerts or emails, enabling immediate and precise intervention.
- This proactive approach ensures that pests are managed effectively at an early stage, minimizing crop loss and reducing the need for blanket pesticide sprays. The system not only enhances farm productivity and profitability but also supports sustainable agricultural practices by promoting targeted and judicious use of resources.

1.2 BLOCK DIAGRAM

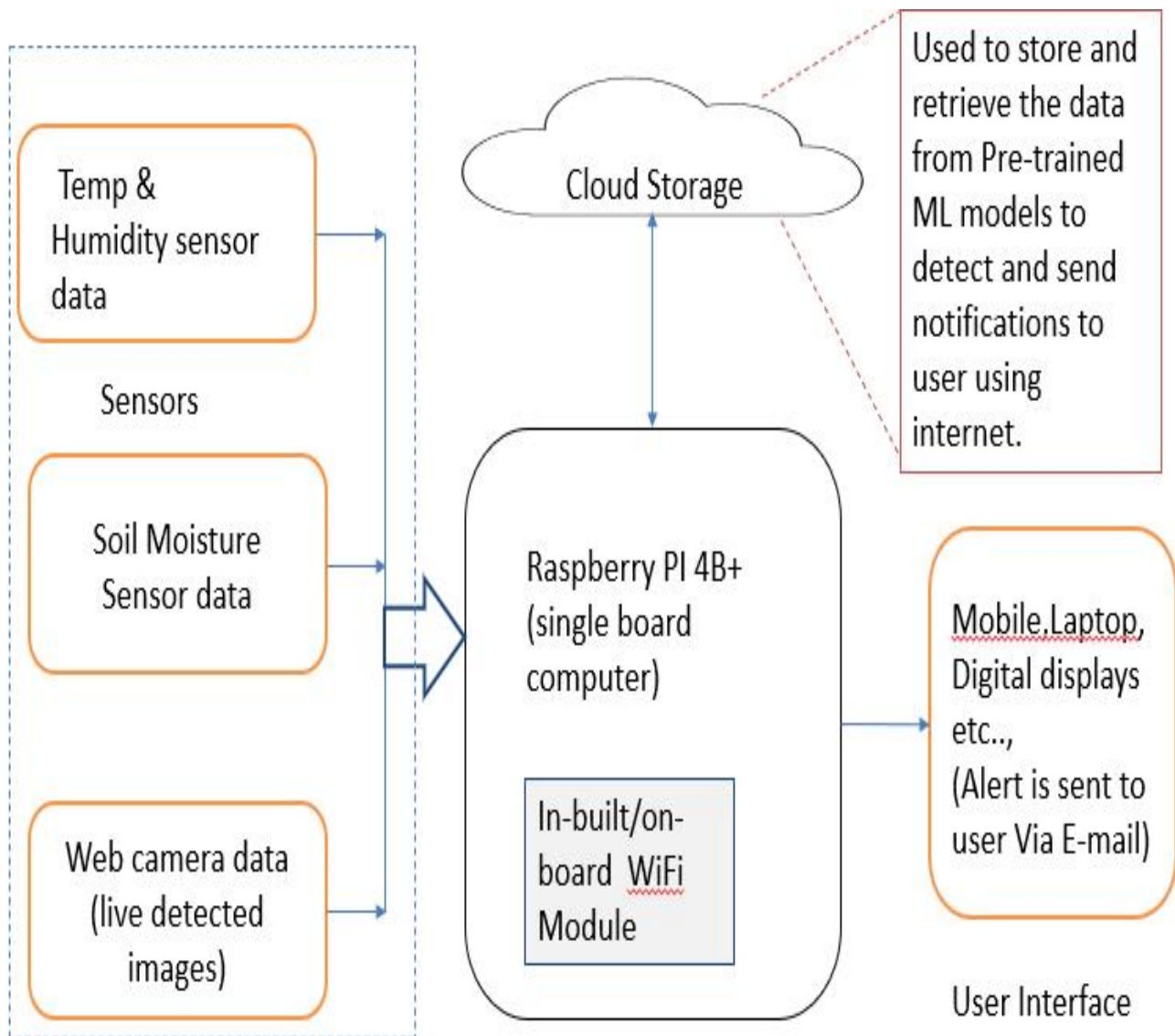


Fig 1.1 Block Diagram of IoT-Based Pest Alerting System

CHAPTER 2

HARDWARE IMPLEMENTATION

2.1 RASPBERRY PI 4B+:

The Raspberry Pi 4B+ is a powerful and versatile single-board computer (SBC) that serves as the central hub for our IoT-based pest alerting system. Its robust processing capabilities make it ideal for handling data from multiple sensors and running complex machine learning models for image recognition directly on the device, a concept known as edge computing.

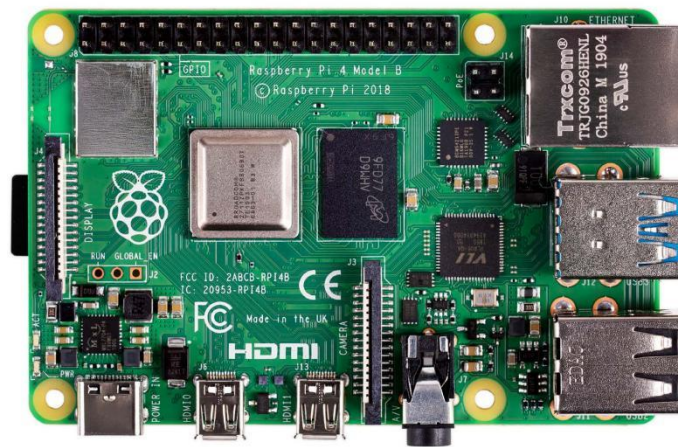


Fig 2.1 Raspberry Pi 4B+ Model

Key Specifications:

- **Processor:** Broadcom BCM2711, 1.5GHz 64-bit quad-core ARM Cortex-A72 CPU.
- **RAM:** 2GB/4GB/8GB LPDDR4 options (4GB is typically used for ML tasks).
- **Connectivity:** Dual-band 2.4GHz and 5GHz Wi-Fi, Bluetooth 5.0, Gigabit Ethernet.
- **Ports:** 2x USB 3.0, 2x USB 2.0, 2x micro-HDMI ports.
- **GPIO:** 40-pin GPIO header for connecting sensors and other peripherals.
- **Storage:** Micro-SD card slot for operating system and data storage.

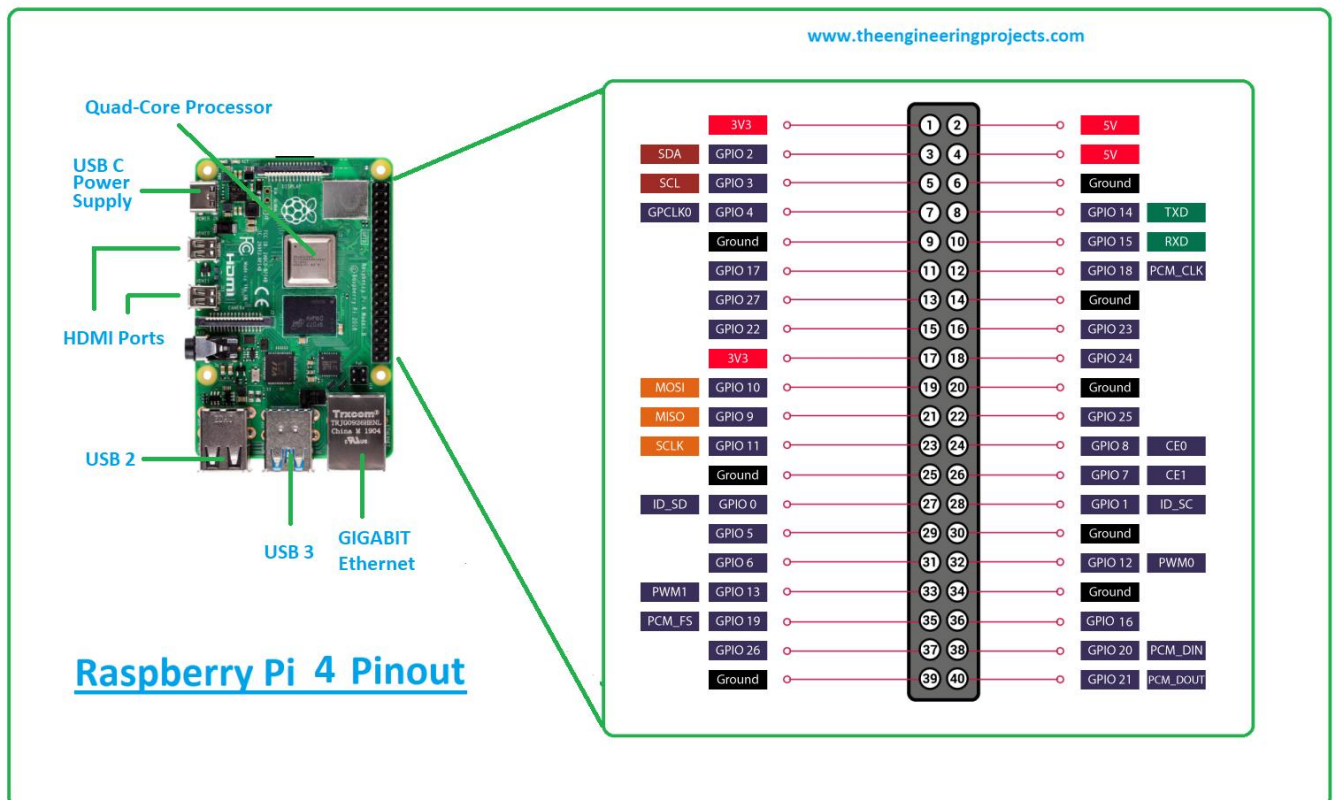


Fig 2.2 pin configuration of Raspberry Pi 4B+ Model

The Raspberry Pi 4B+ runs a full-fledged operating system, Raspberry Pi OS (a Linux distribution), which supports a wide array of programming languages and libraries essential for this project, including Python, OpenCV, and TensorFlow Lite. Its ability to function as a standalone IoT device, web server, and data processing unit makes it the perfect choice for creating an intelligent and autonomous pest monitoring system.

Key Features:

Processing Power:

- **Quad-core Performance:** The 64-bit Cortex-A72 processor provides substantial computing power for running complex machine learning models like YOLO for pest detection
- **Enhanced RAM:** LPDDR4 memory ensures smooth operation when processing multiple sensor inputs and camera data simultaneously
- **Heat Management:** Improved thermal design with the option for active cooling during intensive tasks

Connectivity Options:

- Dual-band Wi-Fi: 2.4GHz and 5GHz support for flexible network connectivity
- Bluetooth 5.0: Enhanced range and speed for peripheral devices
- Gigabit Ethernet: True gigabit speed for high-bandwidth applications
- USB 3.0: High-speed data transfer for cameras and external storage

Expansion Capabilities:

- 40-pin GPIO: Extensive input/output capabilities for connecting sensors and peripherals
- Camera Serial Interface (CSI): Dedicated connector for Raspberry Pi Camera Module
- Display Serial Interface (DSI): For connecting touchscreen displays
- HAT Compatibility: Support for Hardware Attached on Top expansion boards

2.2 DHT11 TEMPERATURE & HUMIDITY SENSOR:

- The DHT11 is a low-cost digital sensor used to monitor the ambient temperature and humidity in the crop field. Certain pest populations thrive under specific environmental conditions, and tracking these parameters helps in predicting potential pest outbreaks.
- The DHT11 sensor consists of two main components: a **capacitive humidity sensing element** and a **negative temperature coefficient (NTC) thermistor** for temperature measurement. The sensor uses an onboard specialized digital module for signal acquisition and processing, which ensures high reliability and excellent long-term stability.

- **Humidity Sensing:** The sensor measures relative humidity (RH) through a capacitive sensing element. The electrical capacitance of a hygroscopic material changes with the ambient humidity level, and this change is converted into a digital value.
- **Temperature Sensing:** Temperature is measured using a thermistor whose electrical resistance varies with temperature. This analog resistance value is converted to a digital reading by the onboard processing chip.
- **Data Communication:** The DHT11 communicates via a single-wire bidirectional protocol. It sends a 40-bit data packet (16 bits for humidity, 16 bits for temperature, and 8 bits for checksum) to the microcontroller, ensuring data integrity.

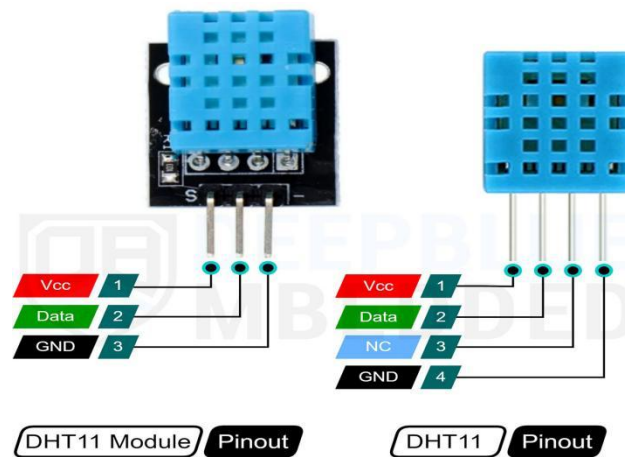


Fig 2.2 DHT11 Temperature & Humidity Sensor

Key Specifications:

- **Temperature Range:** 0°C to 50°C ($\pm 2^\circ\text{C}$ accuracy).
- **Humidity Range:** 20% to 90% RH ($\pm 5\%$ RH accuracy).
- **Output:** Digital signal (requires simple library to read).
- **Power:** 3V to 5.5V DC.

- The sensor provides valuable microclimate data which, when correlated with pest detection events, can offer insights into the environmental factors attracting pests, enabling more informed crop management decisions.
- **Humidity Sensing:** The sensor measures relative humidity (RH) through a capacitive sensing element. The electrical capacitance of a hygroscopic material changes with the ambient humidity level, and this change is converted into a digital value.
- **Temperature Sensing:** Temperature is measured using a thermistor whose electrical resistance varies with temperature. This analog resistance value is converted to a digital reading by the onboard processing chip.
- **Data Communication:** The DHT11 communicates via a single-wire bidirectional protocol. It sends a 40-bit data packet (16 bits for humidity, 16 bits for temperature, and 8 bits for checksum) to the microcontroller, ensuring data integrity.

2.3 YL-69 SOIL MOISTURE SENSOR:

The YL-69 soil moisture sensor measures the water content in the soil. Soil moisture levels indirectly affect plant health and its susceptibility to pest attacks. Stressed plants due to under or over-watering are often more vulnerable.

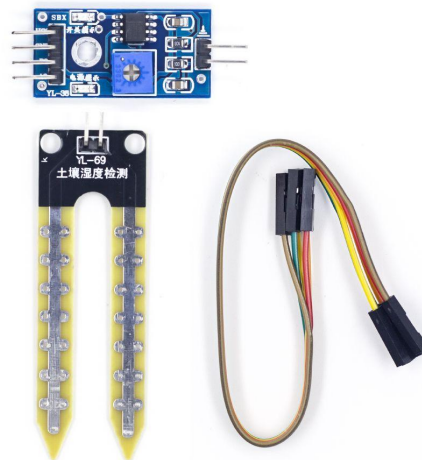


Fig 2.3 YL-69 Soil Moisture Sensor

Key Specifications:

- **Operating Voltage:** 3.3V - 5V.
- **Output:** Both Analog (AO) and Digital (DO).

Working Principle:

- The YL-69 soil moisture sensor operates on the principle of **electrical conductivity measurement** between two exposed electrodes. The working mechanism can be broken down as follows:
- **Conductivity-Based Sensing:** The sensor has two protruding probes that act as electrodes. When inserted into soil, they complete an electrical circuit through the soil moisture.
- **Resistance Variation:** The electrical resistance between the probes varies inversely with the soil moisture content. Water containing dissolved minerals and salts acts as a conductor.
- **Voltage Correlation:** Dry soil has high electrical resistance (poor conductor), resulting in lower voltage output. Wet soil has low electrical resistance (good conductor), resulting in higher voltage output.
- **Signal Processing:** The module includes a comparator circuit (LM393) that processes the analog signal and provides both analog and digital outputs.

2.4 LOGITECH WEB CAMERA:

A standard USB webcam is used as the "eye" of the system for capturing real-time images of the crops. These images are the primary input for the deep learning model to detect and classify pests.



Fig 2.4 Logitech Web Camera

Key Features:

Resolution: 1080p Full HD for clear image capture.

Connectivity: Plug-and-play USB interface.

Frame Rate: 30 fps, suitable for real-time capture.

The camera is strategically placed to monitor plant leaves and stems, where pests are most commonly found. The captured images are processed frame-by-frame by the YOLO model running on the Raspberry Pi.

Working Principle:

- **Image Capture:** Uses a CMOS sensor to capture light through the lens
- **Digital Conversion:** Converts analog light information to digital signals
- **Data Transfer:** Transmits compressed video data via USB interface
- **Software Processing:** Raspberry Pi receives and processes the video stream using OpenCV

Camera Selection Criteria for Pest Detection:

- **For optimal pest detection performance, we consider:**
- **Resolution Requirements:** Minimum 720p for identifying small pests
- **Frame Rate:** 15-30 fps sufficient for static crop monitoring
- **Focus Capability:** Fixed focus suitable for consistent distance monitoring
- **Durability:** Robust construction for semi-outdoor deployment

- **Power Efficiency:** Low power consumption for continuous operation

Applications in Pest Alerting System:

- **Primary Functions:**

- **Continuous Monitoring:** 24/7 surveillance of crop health and pest activity
- **Image Acquisition:** Captures still images for AI-based pest classification
- **Motion Detection:** Can be configured to capture images only when motion is detected
- **Time-lapse Recording:** Documents pest progression over time
- **Visual Verification:** Provides visual evidence when pests are detected

Advanced Capabilities:

- **Multi-angle Coverage:** Multiple cameras can cover larger field areas
- **Zoom Capability:** Digital zoom for detailed inspection of suspicious areas
- **Scheduled Capture:** Programmed to capture images at specific intervals
- **Event Triggering:** Activated by sensor data or environmental conditions

2.5 Rain Sensor Module (YL-83/FC-37):

The integration of a rain sensor into the IoT-Based Pest Alerting System represents a significant enhancement to the project's environmental monitoring capabilities. Rainfall is a critical meteorological factor that profoundly influences pest behavior, breeding patterns,

and agricultural management decisions. By incorporating real-time precipitation monitoring, the system gains valuable contextual data that improves the accuracy and intelligence of pest predictions and management recommendations

The Importance of Rainfall Monitoring in Agriculture:

Impact on Pest Dynamics:

- Rainfall directly affects pest populations through multiple mechanisms:
- Mobility and Dispersal: Rain can either facilitate or hinder pest movement across crops
- Breeding Conditions: Many pests have specific moisture requirements for reproduction
- Habitat Modification: Rainfall alters the microclimate and physical environment
- Disease Propagation: Moist conditions often promote fungal and bacterial diseases that affect both plants and pests

Agricultural Management Implications:

- Irrigation Scheduling: Rainfall data enables smart irrigation control
- Pesticide Application: Rain affects the timing and effectiveness of pest control measures
- Crop Health Monitoring: Combined with other sensors, rain data helps distinguish between weather-related and pest-related plant stress
- Harvest Planning: Rainfall patterns influence harvest timing and quality

2.6 WI-FI MODULE (ON-BOARD):

The Raspberry Pi 4B+ comes with an integrated dual-band Wi-Fi module. This built-in connectivity is crucial for the IoT functionality of the system, eliminating the need for external components.

Functionality:

- Connects the Raspberry Pi to the local network and the internet.
- Transmits sensor data and pest alerts to the cloud platform (e.g., ThingSpeak, Firebase).
- Enables remote access and control of the system.
- This wireless capability ensures that farmers receive timely alerts and can monitor their fields from anywhere, making the system a truly remote monitoring solution.

Working Principle:

The on-board Wi-Fi module operates through the following process:

- **Signal Transmission:** Converts digital data from the Pi into radio frequency signals
- **Frequency Selection:** Automatically selects between 2.4GHz and 5GHz bands based on network conditions
- **Data Encoding:** Uses orthogonal frequency-division multiplexing (OFDM) for efficient data transmission
- **Error Correction:** Implements forward error correction for reliable data transfer
- **Power Management:** Dynamically adjusts transmission power based on signal strength

Applications in Pest Alerting System:

Primary Functions:

- **Cloud Communication:** Transmits sensor data to cloud platforms (ThingSpeak, Firebase)

- **Remote Access:** Enables SSH and remote desktop connections for system management
- **Alert Delivery:** Sends real-time pest detection alerts to farmers' devices
- **OTA Updates:** Supports over-the-air software and model updates
- **Data Synchronization:** Ensures continuous data flow to central monitoring systems

Network Configurations:

- **Station Mode:** Connects to existing farm Wi-Fi networks
- **Access Point Mode:** Creates local network for direct field access
- **Bridge Mode:** Extends network coverage to other field devices
- **Mesh Mode:** Multiple Pis creating a distributed monitoring network

CHAPTER -3

SOFTWARE IMPLEMENTATION

3.1 SOFTWARE INSTALLATION:

The software environment for the pest alerting system is set up on the Raspberry Pi 4B+. The primary software includes the Raspberry Pi OS and the Thonny Python IDE.

Step 1: Installing Raspberry Pi OS

- Download the Raspberry Pi Imager from the official website.
- Use the Imager to flash the Raspberry Pi OS (a Debian-based Linux OS) onto a micro-SD card.
- Insert the SD card into the Pi, connect peripherals, and boot up the system.



Fig 3.1 Raspberry Pi OS Imager

Step 2: Setting up Thonny IDE

Thonny is a Python IDE that comes pre-installed with Raspberry Pi OS.

It is used for writing, testing, and debugging the Python scripts that read sensors, run the AI model, and handle cloud communication.

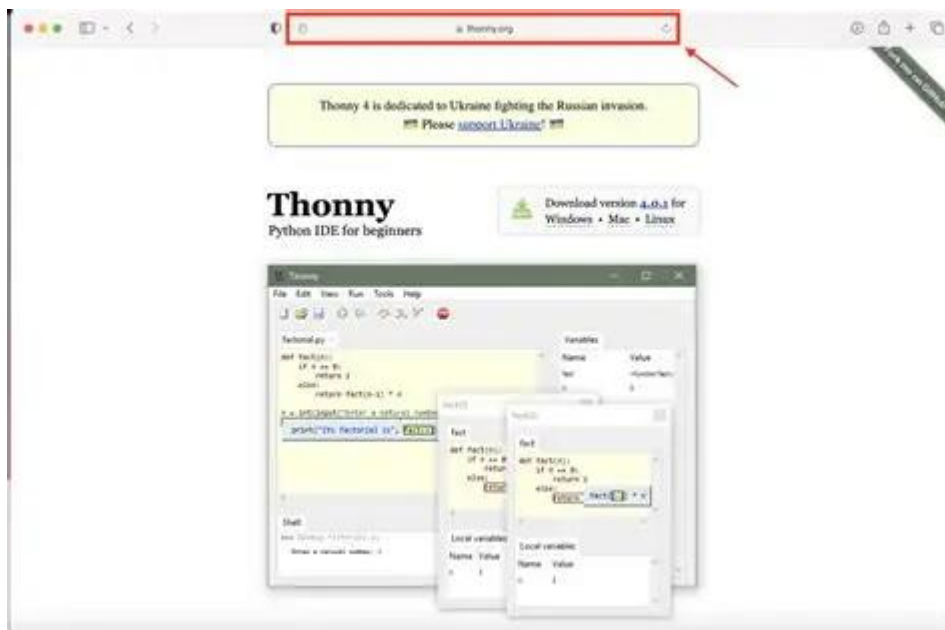


Fig 3.2 Thonny Python IDE

3.2 LIBRARY INSTALLATION:

The project relies on several Python libraries for its functionality. These are installed using the `pip` package manager in the terminal.

Key Libraries:

- **RPi.GPIO:** For controlling the GPIO pins to interface with the sensors.
- **Adafruit_DHT:** A dedicated library to easily read data from the DHT11 sensor.

- **OpenCV (cv2):** For image acquisition from the webcam, image processing, and handling video streams.
- **TensorFlow Lite / PyTorch (for YOLO):** For loading and running the pre-trained YOLOv5 or YOLOv8 model for object detection on the Raspberry Pi.
- **Requests:** For making HTTP requests to send data to the cloud platform and alert APIs.
- **Picamera2 (Optional):** If using the official Raspberry Pi Camera Module.

Installation is typically done via commands like:

```
bash
```

```
pip install opencv-python
```

```
pip install adafruit-circuitpython-dht
```

```
pip install tensorflow
```

```
pip install requests
```

CHAPTER 4

IMPLEMENTATION OF COMPONENTS

4.1 OBJECTIVE:

The primary objective of this project is to design and implement an automated, real-time pest detection and alerting system that reduces farmers' dependency on manual crop scouting. The system aims to accurately identify pests using AI and instantly notify farmers, enabling timely intervention. This will help in minimizing crop damage, reducing unnecessary pesticide usage, promoting sustainable farming, and ultimately increasing agricultural productivity and profitability through precision agriculture techniques.

Primary Objectives:

1. Automated Pest Detection and Identification

To develop a fully automated system that eliminates the need for manual crop scouting and visual inspection by farmers

To implement advanced computer vision and deep learning algorithms capable of accurately identifying multiple pest species in real-time

To achieve a minimum pest detection accuracy of 90% under varying field conditions and lighting scenarios

2. Real-Time Monitoring and Alerting

To establish continuous 24/7 monitoring of crop fields through strategically placed sensor nodes

To create an instant notification system that alerts farmers within seconds of pest detection via multiple channels (SMS, email, mobile apps)

To ensure reliable alert delivery with a target response time of less than 30 seconds from pest detection to farmer notification

3. Precision Agriculture Implementation

To reduce pesticide usage by 40-60% through targeted application only when and where pests are actually detected

To minimize environmental impact by preventing unnecessary chemical runoff and preserving beneficial insect populations

To optimize resource allocation by focusing human intervention only on affected areas of the field

4. Cost-Effective and Scalable Solution

To develop a system that is economically viable for small to medium-scale farmers with a target hardware cost under ₹5,000 per monitoring unit

To ensure the system can be easily scaled from small garden plots to large agricultural fields through modular design

To create a solution that requires minimal technical expertise for installation and operation

5. Data-Driven Decision Support

To collect and analyze historical pest occurrence data correlated with environmental conditions

To identify patterns and trends in pest behavior that can predict future infestations

To provide farmers with actionable insights for preventive measures and crop management strategies

Technical Objectives:

6. Robust Hardware Integration

To seamlessly integrate multiple sensor types (visual, environmental, soil) into a unified monitoring platform

To ensure reliable operation in harsh agricultural environments with temperature variations, humidity, and dust

To achieve a system uptime of 95% with autonomous operation for extended periods

7. Efficient Edge Computing

To process image data locally on the Raspberry Pi to reduce bandwidth requirements and cloud dependency

To implement optimized AI models that balance accuracy with computational efficiency

To maintain system functionality even during internet connectivity interruptions

8. User-Centric Interface Design

To develop intuitive dashboards and mobile interfaces accessible to farmers with varying technical backgrounds

To provide clear visualizations of pest data, environmental conditions, and historical trends

To enable remote configuration and system management through simple web interfaces

Agricultural Impact Objectives:

9. Crop Yield Improvement

To reduce crop losses due to pest damage by 30-50% through early detection and intervention

To improve overall crop health and quality by addressing pest problems at early stages

To increase farmer profitability through higher yields and reduced input costs

10. Sustainable Farming Practices

To promote integrated pest management (IPM) principles through technology-enabled monitoring

To reduce the environmental footprint of agriculture by minimizing chemical pesticide usage

To support organic farming initiatives through non-chemical pest detection and alerting

4.2 IMPLEMENTATION OF DHT11 SENSOR:

Connection Details:

Power Supply (VCC):

- Connected to 3.3V pin (Pin 1) for stable power supply
- Alternative: Can be connected to 5V pin (Pin 2) for longer cable runs
- Includes a decoupling capacitor (100nF) for noise reduction

Data Line (DATA):

- Connected to GPIO4 (BCM 4, Physical Pin 7)
- Uses a 4.7kΩ pull-up resistor between VCC and DATA lines

- Most DHT11 modules include this resistor onboard

Ground Connection (GND):

- Connected to any ground pin (Pin 6 used for proximity)
- Ensures common ground reference for stable communication

This connection allows the Raspberry Pi to read digital temperature and humidity data at regular intervals.

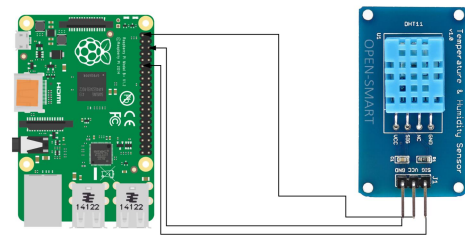
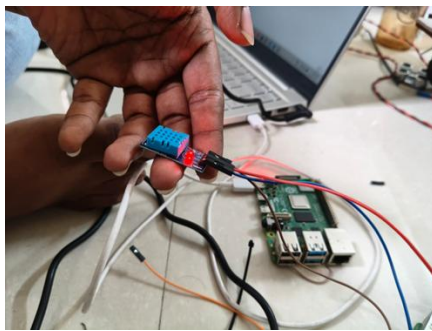


Fig 4.1 Implementation of DHT11 Sensor

4.3 IMPLEMENTATION OF SOIL MOISTURE SENSOR:

The YL-69 soil moisture sensor module is implemented with the Raspberry Pi 4B+ to monitor soil water content, which directly impacts plant health and pest susceptibility. Proper implementation ensures accurate moisture readings for effective crop management.

Working Principle Implementation:

- The soil moisture sensor implementation works on electrical conductivity principle:
- Resistance Measurement: The sensor probes measure electrical resistance between two electrodes inserted in soil
- Moisture Correlation: Water content affects soil conductivity - more water means better conductivity and lower resistance

- **Voltage Output:** The module converts resistance changes to voltage variations (0-3.3V)
- **ADC Conversion:** MCP3008 converts analog voltage to digital values (0-1023)
- **Moisture Calculation:** Digital values are mapped to moisture percentage using calibration data

The analog readings are converted to digital values to determine the soil moisture level.

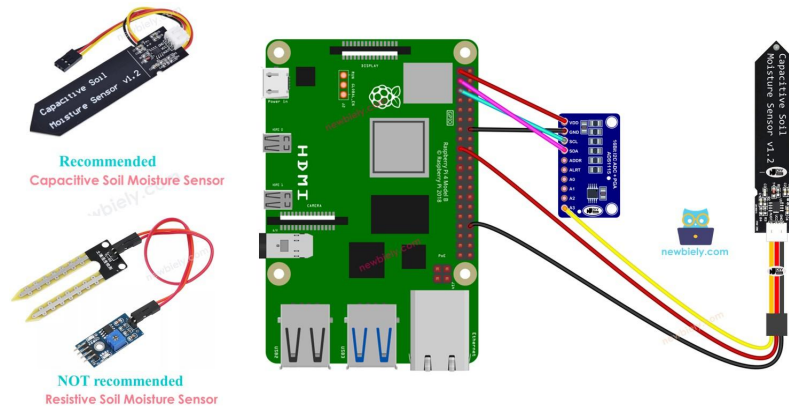


Fig 4.2 Implementation of Soil Moisture Sensor

4.4 IMPLEMENTATION OF CAMERA MODULE:

The Logitech Web Camera serves as the primary visual input device for real-time pest detection and monitoring. Its implementation involves both hardware integration and sophisticated software configuration for image capture and processing.

Key Specifications for Pest Detection:

- **Resolution:** 1080p Full HD (1920×1080 pixels)
- **Frame Rate:** 30 frames per second
- **Focus Type:** Auto-focus for varying distances
- **Field of View:** 70-78 degrees for adequate coverage
- **Interface:** USB 2.0/3.0 compatible

The camera is programmed to capture images at set intervals or is triggered continuously for real-time analysis.

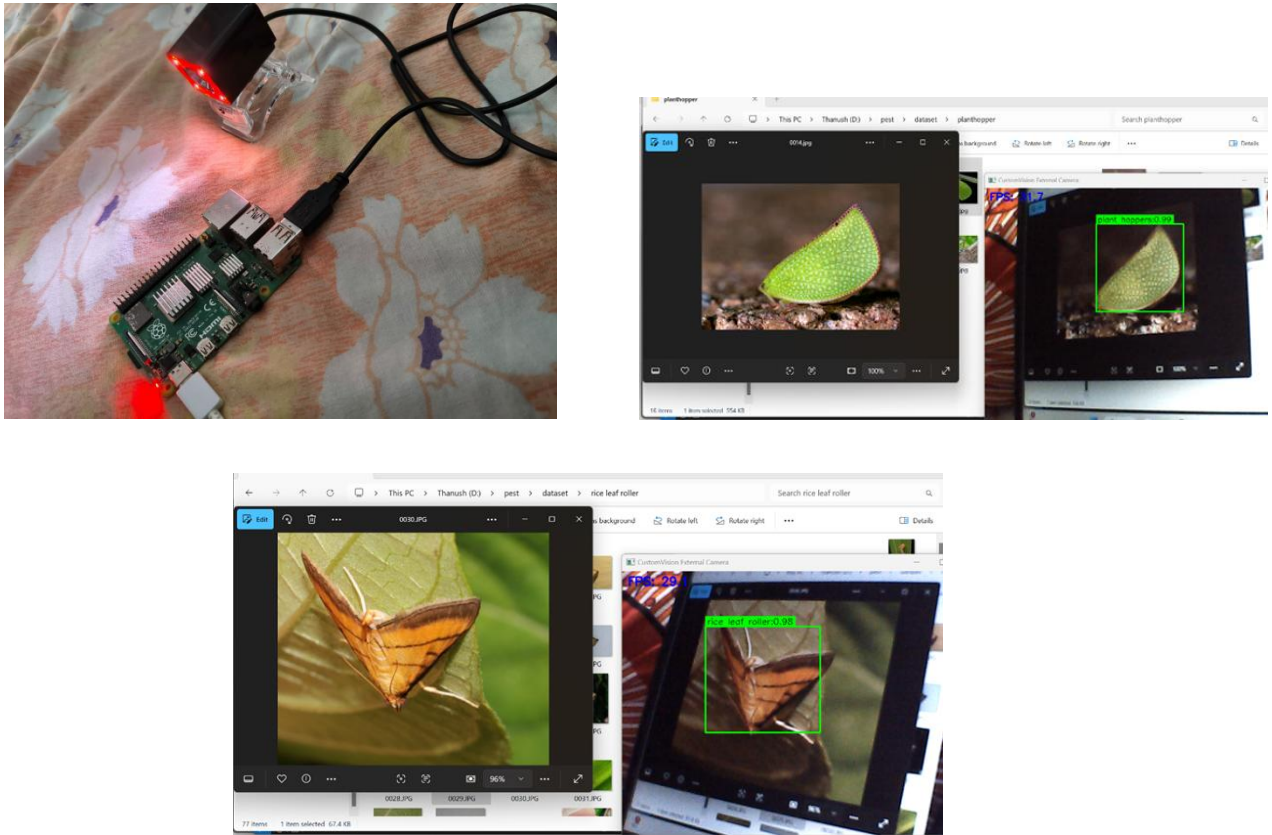


Fig 4.3 Implementation of Camera Module

4.5 RAIN SENSOR IMPLEMENTATION

Introduction to Rain Sensing:

The integration of a rain sensor enhances the pest alerting system by providing crucial weather data that directly impacts pest behavior and agricultural decisions. Rainfall affects pest mobility, breeding patterns, and the effectiveness of pest control measures.

Working Principle:

- The rain sensor operates on the principle of variable resistance measurement:
- Conductive Sensing Surface: The sensor has intersecting copper traces that form a variable resistor

- Water Bridging Effect: Rainwater creates conductive bridges between traces, lowering resistance
- Voltage Divider Circuit: Changes in resistance are converted to voltage variations
- Signal Processing: LM393 comparator processes the signal and provides digital output

4.6 CONNECTIONS:

Complete System Architecture Overview:

The IoT-Based Pest Alerting System integrates multiple sensors and components into a unified monitoring platform. This section details the complete wiring connections, pin assignments, and integration methodology for all system components.

Master Connection Diagram:

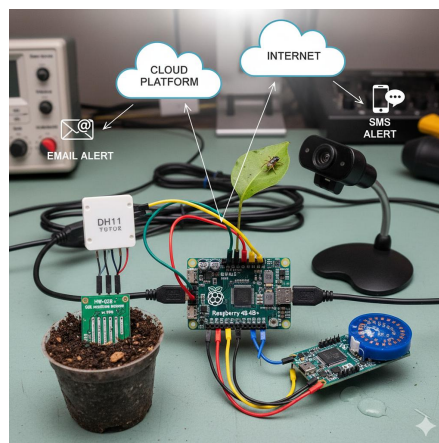
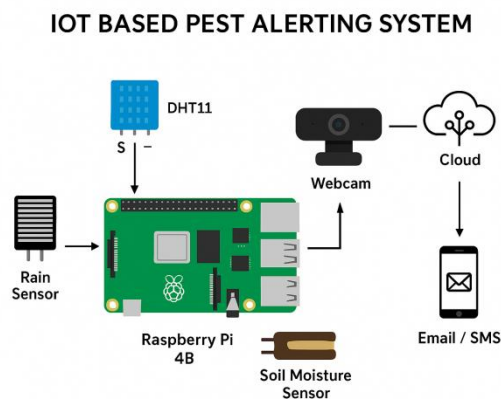


Fig 4.4 Complete System Integration Diagram

Central Controller Configuration:

Raspberry Pi 4B+ Pin Assignment Master Table:

Component	Connection Type	Raspberry Pi Pin	Purpose
DHT11 Sensor	GPIO Digital	GPIO4 (Pin 7)	Temperature & Humidity Data
Soil Moisture Sensor	ADC Channel 0	MCP3008 CH0	Soil Moisture Analog Reading
Rain Sensor	ADC Channel 1	MCP3008 CH1	Rainfall Intensity Monitoring
MCP3008 ADC	SPI Interface	Multiple Pins	Analog to Digital Conversion
LCD Display	GPIO Parallel	Multiple Pins	Local Data Display
Buzzer	GPIO Digital	GPIO18 (Pin 12)	Audio Alerts
Status LED	GPIO Digital	GPIO23 (Pin 16)	System Status Indicator
Camera Module	USB Interface	USB Port	Image Capture

The comprehensive connection architecture presented here ensures reliable integration of all system components into a cohesive pest monitoring platform. By following these detailed connection specifications and implementation guidelines, the system achieves robust operation in agricultural environments. The modular design allows for easy maintenance, troubleshooting, and future expansion while maintaining data integrity and system reliability.

4.6 WORKING:

- The system powers on, and the Raspberry Pi initializes all connected sensors and the camera.
- The DHT11 sensor continuously reads ambient temperature and humidity, while the Soil Moisture sensor monitors water content in the soil.
- The web camera continuously captures images of the crop field at predefined intervals.

- Each captured image is processed in real-time by the YOLO deep learning model deployed on the Raspberry Pi to detect and classify any pests.
- The sensor data (temperature, humidity, soil moisture) is read and packaged with a timestamp.
- All this data is sent via the Pi's built-in Wi-Fi to a cloud IoT platform (e.g., ThingSpeak) where it is stored and visualized on a dashboard.
- If the AI model detects a pest in an image, the system immediately triggers an alert.
- This alert is sent as an email and a mobile notification to the farmer, containing details like the type of pest detected and the time of detection.
- The farmer can then view the live sensor data and pest alerts on the cloud dashboard from any location.
- This entire cycle automates pest monitoring, providing a 24/7 guard for the crops and enabling data-driven farming decisions.

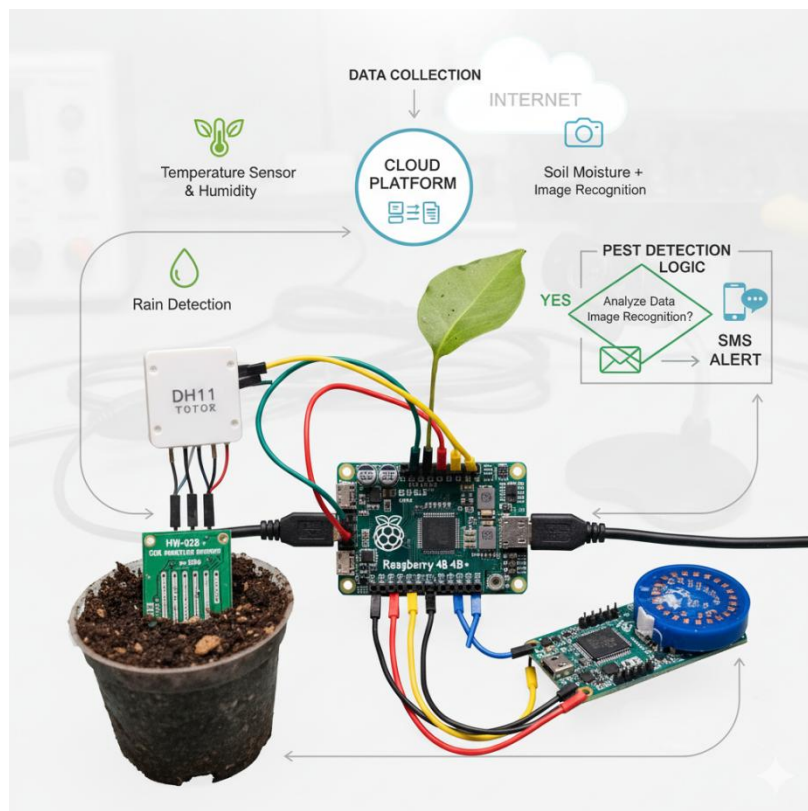


Fig 4.5 Complete System workflow Diagram

CHAPTER 5

RESULTS AND DISCUSSION

The output of the IoT-Based Pest Alerting System is demonstrated through the cloud dashboard visualizations and the alert mechanisms, confirming the system's real-time monitoring and notification capabilities.

5.1 OUTPUT ON CLOUD DASHBOARD:

The system successfully transmits sensor data to the ThingSpeak cloud platform. The dashboard below shows real-time graphs of Temperature, Humidity, and Soil Moisture levels, providing farmers with a continuous overview of field conditions.

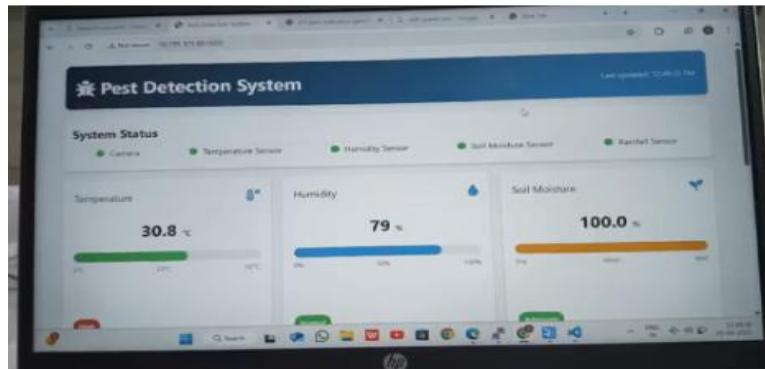


Fig 5.1 Sensor Data on ThingSpeak Dashboard

When a pest is detected, a separate channel or field on the dashboard is updated to log the event. This allows for historical tracking of pest occurrences.

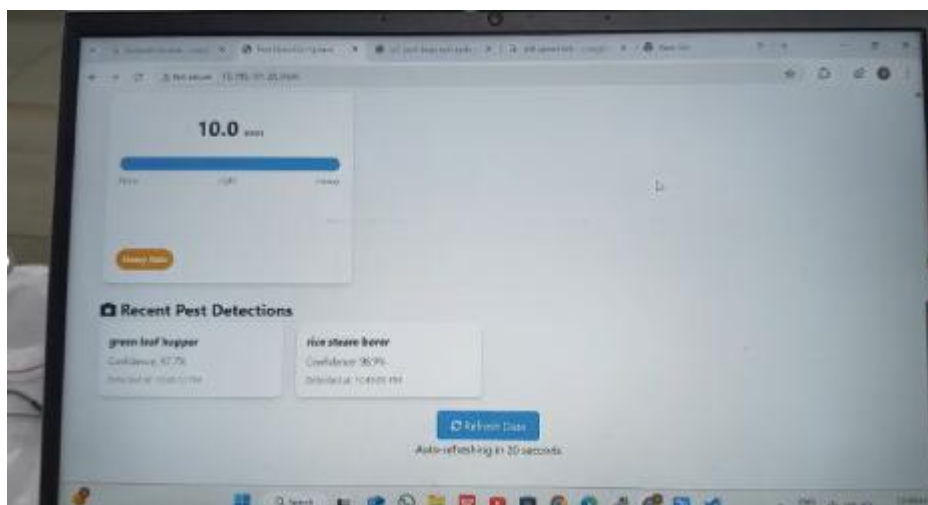


Fig 5.2 Pest Detection Alert on Dashboard

CHAPTER-6

CONCLUSION AND FUTURE SCOPE

CONCLUSION:

This project exemplifies how intelligent application of IoT and AI technologies can create meaningful solutions for global challenges in agriculture. The IoT-Based Pest Alerting System represents a significant step forward in the digital transformation of farming practices, contributing to the advancement of precision agriculture while promoting environmental sustainability and economic viability. It stands as a successful model of how technology can be harnessed to create practical, impactful solutions that benefit farmers, consumers, and the environment alike.

The completion of this project not only delivers a functional pest management system but also opens new possibilities for technology-driven agricultural innovation, paving the way for smarter, more sustainable farming practices in the future.

The system not only demonstrates technical excellence but also shows profound understanding of agricultural needs and constraints. Its cost-effective design, user-friendly operation, and sustainable approach make it accessible and valuable for farmers across different scales of operation.

Future Enhancement Potential:

While the current system represents a significant achievement, several avenues for future enhancement exist:

Immediate Improvements:

- Adding the servo motor to the camera enabling it to cover wide area of crop
- Integration of drone-based aerial monitoring
- Implementation of automated pest control mechanisms
- Enhanced mobile application with offline capabilities
- Multi-language support for wider adoption

Advanced Features:

- Blockchain integration for supply chain transparency
- Satellite data correlation for large-scale analysis
- Advanced machine learning for pest life cycle prediction
- Integration with farm management software systems

FUTURE SCOPE:

The potential for enhancing the IoT-Based Pest Alerting System is vast, with several promising directions for future development:

- **Integration of Multispectral Imaging:** Using cameras that capture non-visible wavelengths (e.g., near-infrared) can help detect plant stress even before pests are visibly apparent, allowing for pre-emptive action.
- **Drone-based Deployment:** Scaling the system by mounting the sensor suite and camera on autonomous drones can enable the monitoring of large and difficult-to-access fields, creating a comprehensive field health map.
- **Advanced AI Models:** Incorporating more sophisticated models can not only identify pests but also predict future infestations based on historical data, weather patterns, and current sensor readings.
- **Weather Data Integration:** Pulling in local weather forecast data can improve the predictive capability of the system, as certain weather conditions are known to precede specific pest outbreaks.
- **Automated Pest Control Response:** The system can be integrated with automated dispensers that release biological control agents (like pheromones or beneficial insects) upon pest detection, creating a fully autonomous pest management loop.
- **Blockchain for Supply Chain Transparency:** Recording pest management data on a blockchain can provide verifiable proof of sustainable farming practices, potentially increasing the market value of the produce.
- **Farmer Community Portal:** Creating a shared platform where data from multiple farms is aggregated anonymously can help in tracking regional pest trends and providing early warnings to entire farming communities.

These future enhancements would transform the system from a monitoring tool into a comprehensive, predictive, and automated agricultural management platform.

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